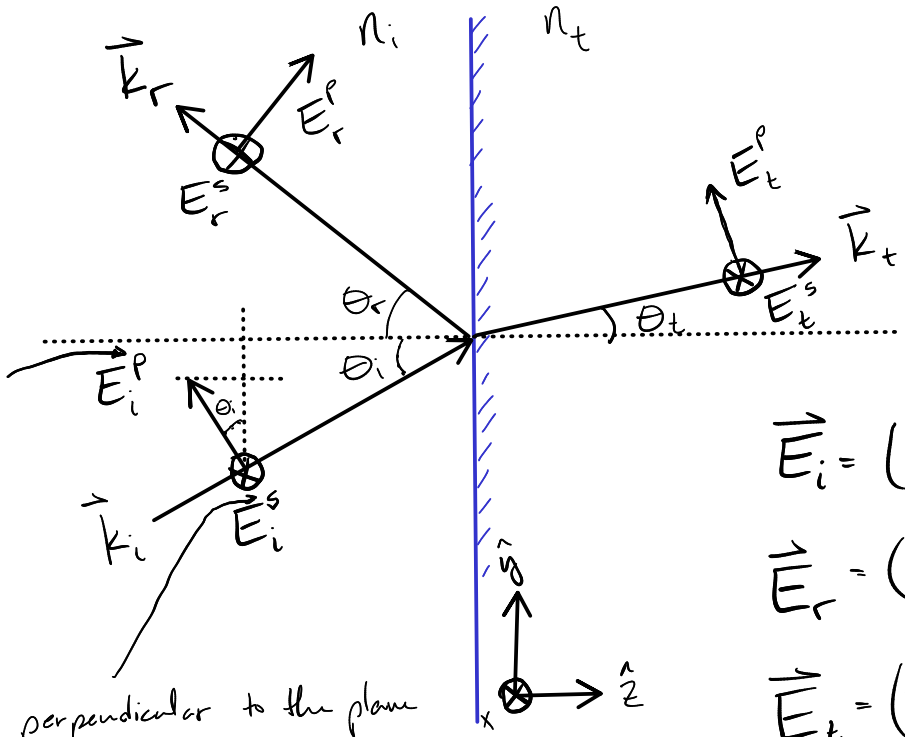
Chapter 3 - Reflection + Refraction

260316 M



$$\vec{k}_i = k_i(\hat{y}\sin\theta_i + \hat{z}\cos\theta_i)$$

$$\vec{k}_r = k_r(\hat{y}\sin\theta_r - \hat{z}\cos\theta_r)$$

$$\vec{k}_t = k_t(\hat{y}\sin\theta_t + \hat{z}\cos\theta_t)$$

"parallel" any vector in the plane is parallel to the plane

perpendicular to the plane "senkrecht"

$$\vec{E}_i = \left(E_i^p(\hat{y}\cos\theta_i - \hat{z}\sin\theta_i) + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)}$$

$$\vec{E}_r = \left(E_r^p(\hat{y}\cos\theta_r + \hat{z}\sin\theta_r) + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$\vec{E}_t = \left(E_t^p(\hat{y}\cos\theta_t - \hat{z}\sin\theta_t) + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

from Maxwell's

↳ Boundary conditions require the components parallel to the interface to be continuous: ($\hat{x}$ + $\hat{y}$ directions in this example)

$$\left(E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s\right)e^{i(k_i(y\sin\theta_i + z\cos\theta_i) - \omega_i t)} + \left(E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s\right)e^{i(k_r(y\sin\theta_r - z\cos\theta_r) - \omega_r t)}$$

$$= \left(E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega_t t)}$$

$\omega_i = \omega_r = \omega_t = \omega$

$$\underbrace{k_i \sin\theta_i = k_r \sin\theta_r}$$ $= k_t \sin\theta_t$

$k = \frac{\omega}{v}$

$v = \frac{c}{n}$

$k_i = \frac{\omega}{v} = \frac{n_i \omega}{c}$ $\quad k_r = \frac{\omega}{v} = \frac{n_i \omega}{c}$

$k_i = k_r$

$\sin\theta_i = \sin\theta_r$

$\boxed{\theta_i = \theta_r}$ law of reflection

$k_i \sin\theta_i = k_t \sin\theta_t$

$\frac{n_i \cancel{\omega}}{\cancel{c}} \sin\theta_i = \frac{n_t \cancel{\omega}}{\cancel{c}} \cdot \sin\theta_t$

$\boxed{n_i \sin\theta_i = n_t \sin\theta_t}$ Snell's Law

$$E_i^p\,\hat{y}\cos\theta_i + \hat{x}E_i^s + E_r^p\,\hat{y}\cos\theta_r + \hat{x}E_r^s = E_t^p\,\hat{y}\cos\theta_t + \hat{x}E_t^s$$

x-direction

$$(1)\quad E_i^s + E_r^s = E_t^s$$

y-direction

$$E_i^p\cos\theta_i + E_r^p\cos\theta_r = E_t^p\cos\theta_t$$

$$\theta_i = \theta_r$$

$$(2)\quad \left(E_i^p + E_r^p\right)\cos\theta_i = E_t^p\cos\theta_t$$

So we have 2 equations but 4 unknowns, so we need the magnetic field equations.

$$\vec{B} = \frac{\vec{k}\times\vec{E}}{\omega} = \frac{n}{c}\,\hat{u}\times\vec{E} \quad \text{where} \quad \hat{u} = \frac{\vec{k}}{|k|}$$

$$(3)\quad n_i\left(E_i^s - E_r^s\right)\cos\theta_i = n_t E_t^s\cos\theta_t$$

$$(4)\quad n_i\left(E_i^p - E_r^p\right) = n_t E_t^p$$

Fresnel Equations

$$r_s = \frac{E_r^s}{E_i^s}$$

$$t_s = \frac{E_t^s}{E_i^s}$$

$$E_i^s + E_r^s = E_t^s \quad \Big| \quad \frac{\cancel{n_i \cos\theta_i}\left(E_i^s - E_r^s\right)}{\cancel{n_i \cos\theta_i}} = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

solve w/ elimination

$$E_i^s - E_r^s = \frac{n_t \cos\theta_t \cdot E_t^s}{n_i \cos\theta_i}$$

$$2E_i^s = \left(\frac{n_t \cos\theta_t}{n_i \cos\theta_i} + 1\right) E_t^s$$

$$2E_i^s = \frac{n_t \cos\theta_t + n_i \cos\theta_i}{n_i \cos\theta_i} \cdot E_t^s$$

$$\boxed{t_s = \frac{E_t^s}{E_i^s} = \frac{2 n_i \cos\theta_i}{n_t \cos\theta_t + n_i \cos\theta_i}}$$

$$\boxed{t_p = \frac{E_t^p}{E_i^p} = \frac{2 n_i \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

$$\boxed{r_s = \frac{E_r^s}{E_i^s} = \frac{n_i \cos\theta_i - n_t \cos\theta_t}{n_i \cos\theta_i + n_t \cos\theta_t}}$$

$$\boxed{r_p = \frac{E_r^p}{E_i^p} = \frac{n_i \cos\theta_t - n_t \cos\theta_i}{n_i \cos\theta_t + n_t \cos\theta_i}}$$

recall:

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \sin\theta_i\right)$$

Reflectance + Transmittance

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Conservation of energy: $P_i = P_r + P_t$ ← power

$$P_i^s = P_r^s + P_t^s$$

$$P_i^p = P_r^p + P_t^p$$

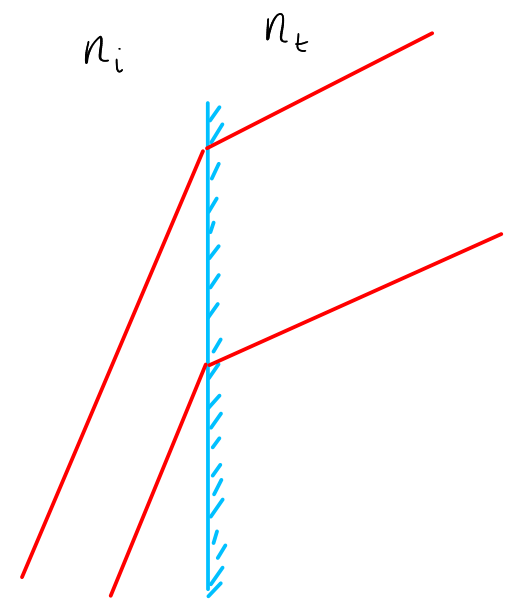
irradiance (intensity)

$$I = \frac{P}{A} \qquad I = c^2 \epsilon_0 \langle E^2 \rangle$$

$$P \propto I \propto E^2$$ ← electric field

reflectance

$$R_s = \frac{P_r^s}{P_i^s} = |r_s|^2 \qquad R_p = \frac{P_r^p}{P_i^p} = |r_p|^2$$



conservation of energy

$$R + T = 1 \Rightarrow \underline{\underline{T = 1 - R}}$$

$$T \neq t^2$$

$$T_s = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_s|^2 \qquad T_p = \frac{n_t \cos\theta_t}{n_i \cos\theta_i} \cdot |t_p|^2$$

Brewster's angle → the angle at which $r_p = 0$

$$\theta_i + \theta_t = \frac{\pi}{2}$$

Snell's → $n_i \sin\theta_i = n_t \sin\theta_t$

$$n_i \sin\theta_i = n_t \underbrace{\sin\left(\frac{\pi}{2} - \theta_i\right)}_{\cos\theta_i}$$ ← at one particular place

$$n_i \sin\theta_B = n_t \cos\theta_B$$ ← Brewster's angle

$$\tan\theta_B = \frac{n_t}{n_i} \Rightarrow \theta_B = \tan^{-1}\left(\frac{n_t}{n_i}\right)$$

Total Internal Reflection

Snell's law: $n_i \sin\theta_i = n_t \sin\theta_t$

solve for θ_t → $\theta_t = \sin^{-1}\left(\frac{n_i}{n_t} \cdot \sin\theta_i\right)$

as this grows

if $n_i / n_t > 1$

if arg $\sin^{-1} > 1$, $\sin^{-1}$ is complex

there is some angle where the light is no longer transmitted → it is reflected!

$\theta_i \to \theta_c = \sin^{-1}\left(\frac{n_t}{n_i}\right)$ $\quad n_i > n_t$

→ $R_s = 1 \quad R_p = 1 \quad \theta_i > \theta_c$

$$r_s = \frac{E_r^s}{E_i^s} = \frac{n_i\cos\theta_i - n_t\cos\theta_t}{n_i\cos\theta_i + n_t\cos\theta_t}$$

so we need a method for calculating $\cos\theta_t$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\cos\theta_t = \sqrt{1-\sin^2\theta_t}$$

substitute using Snell's law

$$r_s = \frac{n_i\cos\theta_i - i n_t\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}{n_i\cos\theta_i + i n_t\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}$$

$$\cos\theta_t = \sqrt{1-\underbrace{\left(\frac{n_i}{n_t}\sin\theta_i\right)^2}_{\text{larger than 1 for total internal reflection}}}$$

book writes:

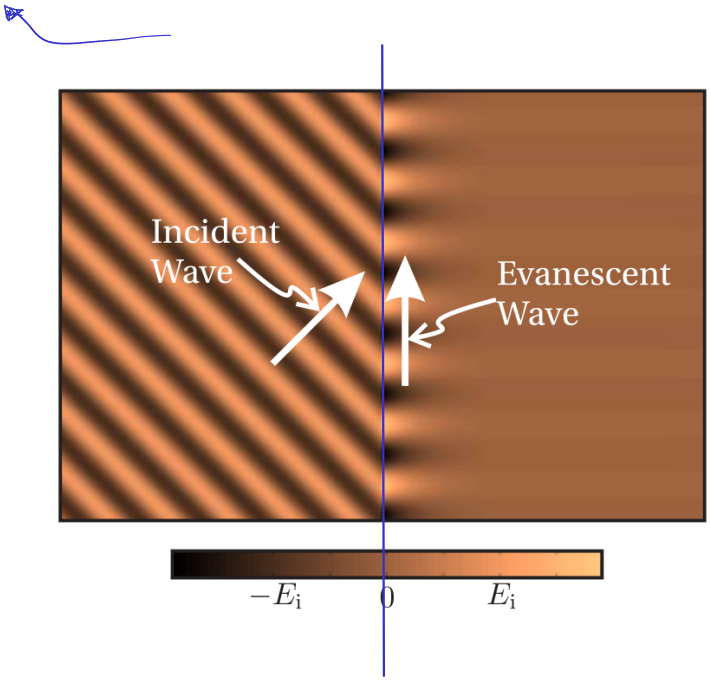
$$\cos\theta_t = i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}$$

only for $n_i > n_t$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\cos\theta_t - \hat{z}\sin\theta_t\right) + \hat{x}E_t^s\right)e^{i(k_t(y\sin\theta_t + z\cos\theta_t) - \omega t)}$$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\,i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1} - \hat{z}\frac{n_i}{n_t}\sin\theta_i\right) + \hat{x}E_t^s\right)e^{i\left(k_t\left(y\frac{n_i}{n_t}\sin\theta_i + iz\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}\right) - \omega t\right)}$$

$$\vec{E}_t = \left(E_t^p\left(\hat{y}\,i\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1} - \hat{z}\frac{n_i}{n_t}\sin\theta_i\right) + \hat{x}E_t^s\right)e^{-k_t z\sqrt{\frac{n_i^2}{n_t^2}\sin^2\theta_i - 1}}\, e^{i\left(k_t y\frac{n_i}{n_t}\sin\theta_i - \omega t\right)}$$



decaying in the z-direction

wave!

Reflections from metal

$$n \rightarrow \tilde{n} = n + ik$$

textbook

$$r_s = \frac{E_r^s}{E_i^s} = \frac{\tilde{n}_i \cos\theta_i - \tilde{n}_t \cos\theta_t}{\tilde{n}_i \cos\theta_i + \tilde{n}_t \cos\theta_t}$$

HW: 3.5, 7, 8, 10, 11, 12, 13, 14

↓ <u>AND</u>

15. When light passes through a prism its path changes once on the way into the prism and then again on the way out. Assume the prism is an equilateral triangle of glass ($n = 1.5$).

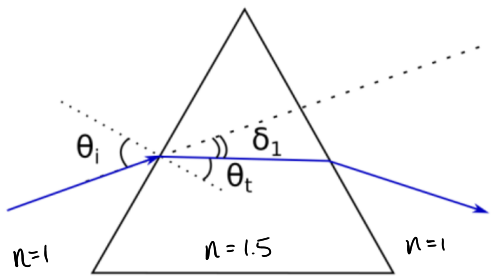
(a) For an exterior incident angle of 30°, what is the transmitted angle in the prism?

$$n_i \sin\theta_i = n_t \sin\theta_t$$

$$\sin\theta_t = \frac{n_i}{n_t}\sin\theta_i \Rightarrow \theta_t = \sin^{-1}\left(\frac{n_i}{n_t}\sin\theta_i\right)$$

$$\theta_t = 19.47^\circ$$

(b) Inside the prism, what is the *deviation* δ_1 of the light's path?

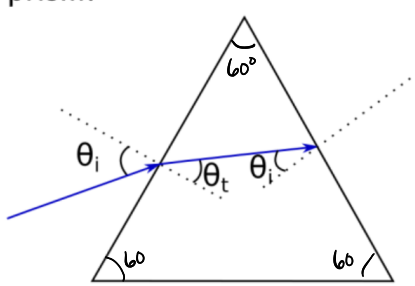


$$\theta_i = \delta_1 + \theta_t$$

$$\delta_1 = \theta_i - \theta_t$$

$$\delta_1 = 30^\circ - 19.47^\circ = 10.53^\circ$$

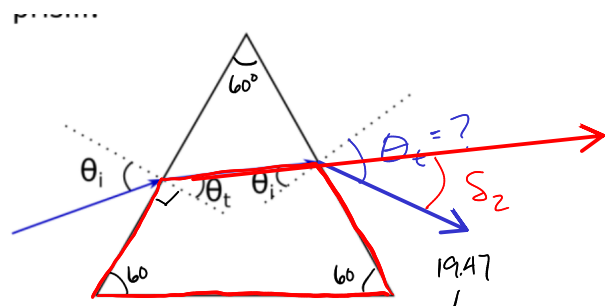
(c) What is the interior incident angle as the light is leaving the prism?



$$90^\circ - 19.47 = 70.53$$

$$180^\circ - 60^\circ - 70.53^\circ = 49.47^\circ$$

$$\theta_i = 90^\circ - 49.47 = \underline{40.53^\circ}$$



19.47

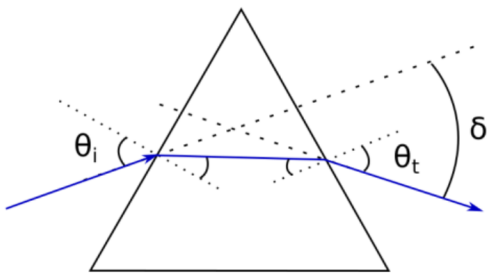
$$360 = 60 + 60 + (90 + \theta_t) + (90 + \theta_i)$$

$$\theta_i = 40.53^\circ$$

(d) What is the final transmitted angle as the light leaves the prism?

(e) What is the second deviation of the light's path?

(f) What is the total deviation of the lights path?



(g) Using excel or google sheets, do this procedure with many exterior incident angles (all integer angles). Don't forget that excel's trig functions only accept radians and inverse trig functions will return angles in radians so you will have to include some conversion factors in the formula (π rad $=$ 180°). Plot the total deviation as a function incident angle and see if there are any interesting features.